

1.Caesar Cipher

Aim

To implement the Caesar Cipher algorithm in C for encryption and decryption of a text message using a specified key value.

Algorithm

Encryption

1. Start.
2. Read the plaintext message.
3. Read the key value.
4. Reduce the key using $\text{key} = \text{key} \% 26$.
5. Read each character of the message:
 - If it is an uppercase letter, shift it forward by the key.
 - If it is a lowercase letter, shift it forward by the key.
 - Leave spaces and special characters unchanged.
6. Display the encrypted text.
7. Apply the reverse shift to decrypt the encrypted text.
8. Display the decrypted text.
9. Stop.

Encryption Formula

For uppercase letters:

$$C = (P - 'A' + \text{key}) \% 26 + 'A'$$

For lowercase letters:

$$C = (P - 'a' + \text{key}) \% 26 + 'a'$$

Decryption Formula

$$P = (C - \text{key} + 26) \% 26$$

PROGRAM :

```
#include <stdio.h>
```

```
void encrypt(char text[], int key) {
```

```
    int i;
```

```

for (i = 0; text[i] != '\0'; i++) {

    if (text[i] >= 'A' && text[i] <= 'Z') {
        text[i] = (text[i] - 'A' + key) % 26 + 'A';
    }

    else if (text[i] >= 'a' && text[i] <= 'z') {
        text[i] = (text[i] - 'a' + key) % 26 + 'a';
    }
}

void decrypt(char text[], int key) {
    int i;

    for (i = 0; text[i] != '\0'; i++) {

        if (text[i] >= 'A' && text[i] <= 'Z') {
            text[i] = (text[i] - 'A' - key + 26) % 26 + 'A';
        }

        else if (text[i] >= 'a' && text[i] <= 'z') {
            text[i] = (text[i] - 'a' - key + 26) % 26 + 'a';
        }
    }
}

int main() {
    char text[100];
    int key;

    printf("Enter the text: ");

```

```
fgets(text, sizeof(text), stdin);
```

```
printf("Enter the key: ");
```

```
scanf("%d", &key);
```

```
key = key % 26;
```

```
encrypt(text, key);
```

```
printf("\nEncrypted Text: %s", text);
```

```
decrypt(text, key);
```

```
printf("Decrypted Text: %s", text);
```

```
return 0;
```

```
}
```

Output

```
D:\CSA5112 - Cryptography\LAB PROGRAMS\caesar_cipher.cpp - [executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Project Clae caesar_cipher.cpp
void encrypt(char text[], int key) {
    int i;
    for (i = 0; text[i] != '\0'; i++) {
        if (text[i] >= 'A' && text[i] <= 'Z') {
            text[i] = (text[i] - 'A' + key) % 26 + 'A';
        }
        else if (text[i] >= 'a' && text[i] <= 'z') {
            text[i] = (text[i] - 'a' + key) % 26 + 'a';
        }
    }
}

void decrypt(char text[], int key) {
    int i;
    for (i = 0; text[i] != '\0'; i++) {
        if (text[i] >= 'A' && text[i] <= 'Z') {
            text[i] = (text[i] - 'A' - key + 26) % 26 + 'A';
        }
        else if (text[i] >= 'a' && text[i] <= 'z') {
            text[i] = (text[i] - 'a' - key + 26) % 26 + 'a';
        }
    }
}

int main() {
    char text[100];
    int key;
    printf("Enter the text: ");
    fgets(text, sizeof(text), stdin);
    printf("Enter the key: ");
    scanf("%d", &key);
    key = key % 26;
    encrypt(text, key);
    printf("Encrypted Text: %s", text);
    decrypt(text, key);
    printf("Decrypted Text: %s", text);
    return 0;
}

Compiler Resources Compile Log Debug Find Results Console
- Errors: 0
- Warnings: 0
- Output Filename: D:\CSA5112 - Cryptography
- Output Size: 324.0166015625 KiB
- Compilation Time: 0.47s
Line: 8 Col: 48 Sel: 0 Lines: 52 Length: 1088 Insert Done parsing in 0.141 seconds
```

```
D:\CSA5112 - Cryptography\Lab
Enter the text: Hello
Enter the key: 3
Encrypted Text: KHOOR
Decrypted Text: Hello

Process exited after 4.435 seconds with return value 0
Press any key to continue . . .
```